

Pakhi Banchalia

Dresden, Germany | pakhibanchalia24@gmail.com | [GitHub](#) | [Portfolio](#)

EDUCATION

- **Technische Universität Dresden** Oct 2025 - Present
Master of Science in Computational Modeling and Simulation (Applied AI track) Dresden, Germany
- **Amrita Vishwa Vidyapeetham Kerala, India** 2021-2025
Bachelor of Technology in Computer Science and Artificial Intelligence Amritapuri, India

RESEARCH EXPERIENCE

- **Aalto University** June 2026 - Present
Summer Research Intern Espoo, Finland
 - Conducting research on social intrinsic motivation and assistance in multi-agent reinforcement learning.
 - Studying information-theoretic approaches to agent autonomy, interaction, and supportive behavior.
- **Lossfunk** July 2025 - Sept 2025
Research Intern Bangalore, India
 - Developed and evaluated homeostatic reinforcement-learning agents in the Crafter environment, using multiple internal variables such as hunger, energy, and safety to shape agent behaviour.
 - Designed experiments to study how alternative reward formulations and competing internal drives affect exploration, survival, behavioural diversity, and policy stability.
 - Compared homeostatic agents against standard reinforcement-learning baselines.
- **Computational Machinery of Cognition (CMC) Lab, TU Dresden** Nov 2024 - Present [Intermittent]
Research Intern (Independent Collaboration) Dresden, Germany
 - Designed RL agents with internal physiological drives to study retention and generalization across changing task distributions, focusing on the relationship between internal regulation and long-term behavioral stability.
 - Implemented a reward-modulated, gradient-free learning algorithm based on weight perturbations to investigate biologically plausible credit assignment without backpropagation.
- **Google Summer of Code, INCF** May 2024 - Sept 2024
Contributor Remote
 - Simulated *C. elegans* neural development using Neural Cellular Automata and evolutionary optimization, representing developmental changes in cell state and connectivity over time.
 - Modeled evolving neural connectivity with Graph Neural Networks and used Topological Data Analysis with Kepler Mapper to inspect developmental and connectivity trajectories.
 - Delivered reproducible, documented open-source research code for computational developmental neuroscience. [\[Code\]](#)

PUBLICATIONS

- Pillai, H.S., **Banchalia, P.**, Jamboji, V., Krishnan, S.R. (2026). *CAN: Continuously Adapting Network*. In *Data Science and Applications, ICDSA 2025. Lecture Notes in Networks and Systems*, vol 1723. Springer, Cham. [\[DOI\]](#)

PROJECTS

- **Alignd: Sponsorship Agent for TinyFish Accelerator**
 - Selected for the TinyFish Accelerator, a selective nine-week programme hosted by TinyFish and Mango Capital for founders building production-oriented agentic web applications.
 - Built an AI agent that discovers and evaluates relevant sponsorship opportunities by navigating live web sources and organizing potential sponsors for further review.
 - Developed a workflow for assessing sponsor relevance and supporting personalised outreach preparation. [\[Code\]](#)
- **SEANN: Self-Expanding and Adaptive Neural Networks**

- Extended the CAN continual-learning framework with dynamic capacity expansion and task-relevant subnetworks.
- Investigated structural plasticity as an alternative to fixed model scaling for reducing interference and catastrophic forgetting.
- **Dopamine: Reward-Modulated Gradient-Free Learning**
 - Implemented reward-modulated weight perturbation as an alternative credit-assignment mechanism and benchmarked early learning performance against gradient-based approaches.
 - Studied whether biologically inspired local updates can produce useful behavior without explicit backpropagation.

TALKS AND POSTERS

- **Brain-like features in homeostatic deep reinforcement learning agents @ Bernstein Conference 2025**
 - Poster presentation on homeostatic RL agents integrating internal and sensory inputs, showing structured behavioral and neural patterns
- **Emergence of Brain-like Features in Homeostatic RL Agents @ BrainBody 2025**
 - Oral presentation; demonstrated coupling between internal states, neural dynamics, and adaptive exploration

TECHNICAL SKILLS

- **Languages:** Python, C++, MATLAB, SQL
- **ML / Research:** PyTorch, TorchVision, Gymnasium (OpenAI Gym), JAX (familiar)
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper (TDA)
- **Research Areas:** Reinforcement Learning, Biologically-Inspired Learning, Agentic AI, Robotics, Neural Cellular Automata, Continual Learning, GNNs
- **Developer Tools:** Git, Linux/Unix, Docker, Weights & Biases (W&B)